Method of Handling Traffic — Crew Instructions

Project Information

- **Speed limit:** 40 mph
- **Road type:** Urban (low-speed)
- **Closure type:** Lane Closure Near Intersection — Undivided
- **Work zone length:** 2,000 ft
- **Jurisdiction:** Thornton
- **Generated:** 2026-08-22

Required Equipment

- 19× Construction/Warning Sign (see sign list below)
  - 1× G20-1 ROAD CONSTRUCTION (NEXT 2000 FT)
  - 1× G20-2 END ROAD WORK
  - 2× G20-5P CONSTRUCTION ZONE plaque
  - 3× R2-10 BEGIN DOUBLE FINES ZONE
  - 3× R2-11 END DOUBLE FINES ZONE
  - 2× R9-9 SIDEWALK CLOSED
  - 2× S1-1 SCHOOL
  - 3× W20-1 ROAD WORK AHEAD
  - 1× W20-5R RIGHT LANE CLOSED AHEAD
  - 1× W4-2R RIGHT LANE ENDS
- 8× Channelizing Drum (36-inch)
- 33× Traffic Cone (28-inch)
- 7× Type C Steady-Burn Warning Light
- 4× Type III Barricade
- 1× Arrow Board
- 1× Portable Light Plant

Sign Placement Schedule

Sign Code	Description	Distance from Taper
S1-1	SCHOOL	800 ft upstream
W20-1	ROAD WORK AHEAD	300 ft upstream
W20-5R	RIGHT LANE CLOSED AHEAD	200 ft upstream
W4-2R	RIGHT LANE ENDS	100 ft upstream
G20-5P	CONSTRUCTION ZONE plaque	Within work zone
G20-2	END ROAD WORK	Downstream of work zone
R2-10	BEGIN DOUBLE FINES ZONE	250 ft upstream (1/2 C inside the outermost sign, per Case 18)
G20-1	ROAD CONSTRUCTION (NEXT 2000 FT)	100 ft upstream of the work zone (within the buffer space)
R2-11	END DOUBLE FINES ZONE	250 ft downstream of the work zone

Cross-Street Signs (per approach leg; stations measured from the intersection curb line)

- **north_leg approach** (40 mph, Urban (low-speed); signalized — signal operation review required before work begins, MUTCD §6N.12 / Part 4): W20-1 at 200 ft (twice the key distance A) and R2-10 at 100 ft (at A) on the approach side — the S-630-1 Sheet 10 advance-signing key gives A = 100 ft for this leg's speed and road type. R2-11 at 100 ft on the departure side.
- **south_leg approach** (35 mph, Urban (low-speed); signalized — signal operation review required before work begins, MUTCD §6N.12 / Part 4): W20-1 at 200 ft (twice the key distance A) and R2-10 at 100 ft (at A) on the approach side — the S-630-1 Sheet 10 advance-signing key gives A = 100 ft for this leg's speed and road type. R2-11 at 100 ft on the departure side.

Setup Procedure

Important: Set up from downstream to upstream (against traffic flow). This ensures you are always working behind protection.

1. Position arrow board at the upstream start of the merging taper (station 2,612 ft), centered in the closed right lane. Set to LEFT ARROW mode (traffic merges left, out of the closed lane). Activate flashing.
2. Place the END ROAD WORK sign (G20-2) at the downstream termination (station -150 ft), right side of road 36 ft from centerline.
3. Place 2 traffic cones in the downstream reopening taper, between station -50 ft (lane edge) and station 0 (lane line) at 25 ft spacing — this taper returns traffic into the reopened right lane past the work zone (MUTCD §6B.08 downstream taper). Then place 26 traffic cones along the closed-lane line through the work zone at 80 ft spacing, starting at station 0 (downstream end of work zone) and working upstream to station 2,000 ft. Offset: 21 ft from centerline. Then continue the closed-lane cone line upstream from the work zone to the downstream curb line of the cross street (station 2,168 ft), and again from the upstream curb line (station 2,232 ft) up to the taper — 5 cones total, with no devices inside the intersection itself. The lane must already be closed where it crosses the intersection (MUTCD §6N.12.12).
4. Place G20-5P WORK ZONE plaque(s) within the work zone at approximately 1,000 ft intervals.
5. Place 8 channelizing drums in the merging taper, starting at station 2,332 ft and working upstream to station 2,612 ft. Space drums at 40 ft intervals. Drums transition from the lane edge (32 ft) to the lane line (21 ft), closing the right lane.
6. Place advance warning signs upstream of the taper:
    - W4-2R at 100 ft upstream of taper start (station 2,712 ft), right side.
    - W20-5R at 200 ft upstream of taper start (station 2,812 ft), right side.

    - W20-1 at 300 ft upstream of taper start (station 2,912 ft), right side.
    - S1-1 at 800 ft upstream of taper start (station 3,412 ft), right side.

7. Place the mainline regulatory and bookend signs, single side, 36 ft from centerline: R2-10 BEGIN DOUBLE FINES ZONE at station 2,862 ft (half the C spacing inside the outermost W20-1, per S-630-1 Case 18); G20-1 at station 2,100 ft; R2-11 END DOUBLE FINES ZONE at station -250 ft (downstream, past the lane reopening).
8. Place the cross-street advance sets — one set per approach leg. Stations for each set are measured along that leg FROM the intersection curb line, increasing away from the intersection:
    - **north_leg approach** (40 mph, Urban (low-speed)): W20-1 ROAD WORK AHEAD at 200 ft and R2-10 BEGIN DOUBLE FINES ZONE at 100 ft, both on the approach side 36 ft from that road's centerline — the W20-1 sits at twice the key distance A, the R2-10 at A, where the S-630-1 Sheet 10 advance-signing key gives A = 100 ft for this leg's own speed and road type. R2-11 END DOUBLE FINES ZONE at 100 ft on the departure side (facing traffic leaving the intersection). This approach is signalized: signal operation must be reviewed before work begins (MUTCD §6N.12; Part 4) — see the audit trail.
    - **south_leg approach** (35 mph, Urban (low-speed)): W20-1 ROAD WORK AHEAD at 200 ft and R2-10 BEGIN DOUBLE FINES ZONE at 100 ft, both on the approach side 25 ft from that road's centerline — the W20-1 sits at twice the key distance A, the R2-10 at A, where the S-630-1 Sheet 10 advance-signing key gives A = 100 ft for this leg's own speed and road type. R2-11 END DOUBLE FINES ZONE at 100 ft on the departure side (facing traffic leaving the intersection). This approach is signalized: signal operation must be reviewed before work begins (MUTCD §6N.12; Part 4) — see the audit trail.

Sequencing note: The MUTCD does not prescribe the order of mainline versus cross-street setup. Whether the cross-street sets go up before, after, or interleaved with the mainline sequence is the Traffic Control Supervisor's determination for this site.

Takedown Procedure

Important: Remove from upstream to downstream (with traffic flow). Remove advance warning signs first, work zone devices last.

1. Remove advance warning signs: S1-1 first, then W20-1, then W20-5R, then W4-2R.
2. Remove taper drums from upstream to downstream.
3. Deactivate and remove arrow board.
4. Remove work zone cones from upstream to downstream.
5. Remove G20-5P plaques.
6. Remove G20-2 END ROAD WORK sign(s).
7. Remove the mainline R2-10 / R2-11 DOUBLE FINES ZONE pair and the G20-1 sign.
8. Remove each cross-street advance set (W20-1, R2-10, R2-11 per approach). The order of mainline versus cross-street takedown is the Traffic Control Supervisor's determination.

Safety Notes

- All personnel in the work zone must wear ANSI Class 2 (daytime) or Class 3 (nighttime) high-visibility apparel.
- Nighttime operation: Ensure all devices meet retroreflectivity requirements. Flagger stations require 500W quartz illumination at minimum 8 ft height per CDOT S-630-1 Sheet 2 General Note 22.
- Closed lane: this plan draws the closure on the RIGHTMOST lane. That is a modeling assumption, not an MUTCD rule — the work location dictates the lane. If the work occupies a different lane, this plan does not apply as drawn.
- Maintain 305 ft longitudinal buffer space between the taper and the work area at all times.
- Maximum speed reduction per sign installation: 15 mph (CDOT S-630-1 Sheet 2 General Note 3).

- This plan was generated by MHT Tool. A certified Traffic Control Supervisor (TCS) must review, approve, and sign before field implementation.
- Reference: CDOT S-630-1, MUTCD 11th Edition Part 6, Colorado Supplement (effective January 18, 2026).

Pedestrian and Bicycle Accommodation

- A pedestrian facility is affected by this closure. 2 R9-9 SIDEWALK CLOSED signs are placed at the closure (MUTCD §6G.10 — the plain R9-9 legend; no alternate-side assertion, which belongs to R9-10, and no detour-routing signs, which require sidewalk geometry this plan does not carry). Sidewalk closure points are barricaded (MUTCD §6C.02). Pedestrians must not be diverted into an open travel lane.
- The TCS verifies pedestrian and bicycle accommodation on site; where a facility is maintained, it must remain accessible and detectable per MUTCD standards.

Emergency Access

- 2 of 3 same-direction travel lanes remain open (21 ft of traversable width) past the closed lane.
- The Traffic Control Supervisor completes and confirms the emergency access plan for this site and coordinates notification of police, fire, and emergency medical responder agencies prior to work (TCS duties per CDOT Standard Specifications 630.11).
- An emergency access plan is a required minimum element of the approved Method of Handling Traffic (CDOT 630.10(a)); this section is draft input for it, not the approved plan.

Site-Specific Notes

- **adjacent_intersection** — No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure. *(MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.))*
- **adjacent_interchange** — No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure. *(MUTCD §6N.16 p. 851 — Interchanges (11th Ed.))*
- **pedestrian_facility** — Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone. *(MUTCD §6C.02 — pedestrian considerations in work zones)*
- **school_zone** — Added 2 S1-1 SCHOOL signs upstream of the advance warning signs (station 3,412 ft, both sides). *(MUTCD §7B.08 — school zone signing in proximity to work zones)*

Night Operation Notes

- **night_taper_lighting** — Added 7 Type C steady-burn warning lights on taper drums for nighttime visibility. *(MUTCD §6L.07 — warning lights on channelizing devices in nighttime work zones)*
- **night_work_area_lighting** — Added 1 portable light plant at the start of the work zone (25 ft past the buffer) to illuminate the work area. *(CDOT M&S §630.05.1 — illumination requirement for nighttime work zones)*
- **night_retroreflective** — All devices, signs, and barricades must be retroreflective. Verify warehouse inventory before deployment. Inspect retroreflectivity at shift start. *(MUTCD §6F.01 + §6K.01 — retroreflectivity requirements for nighttime visibility)*